

- (c) (i) Amount of electricity used by the solar water heater  
 $= 365 \times 4 \times 3000 \times 50\%$  watt-hours  
 $= 2\,190\,000$  watt-hours  
 $= 2190$  kilowatt-hours (shown)
- (ii) Amount of electricity used by the heat pump water heater  
 $= 365 \times 4 \times (200 + 3000 \times 2\%)$  watt-hours  
 $= 379\,600$  watt-hours  
 $= 379.6$  kilowatt-hours

Difference in electricity used  
 $= (2190 - 379.6)$  kilowatt-hours  
 $= 1810.4$  kilowatt-hours  
 $\approx 1800$  kilowatt-hours  
 $\therefore$  Huan is correct.

11. (a) Angle  $BCD = 1.94$  rad because angles in opposite segments, i.e.  $\angle BAD$  and  $\angle BCD$ , are supplementary.

#### EXAM TIP

Recall that angles in opposite segments are supplementary, i.e. they have a sum of  $\pi$ .

- (b) Using Cosine Rule,  
 $BD^2 = AB^2 + AD^2 - 2(AB)(AD) \cos \angle BAD$   
 $= 3.6^2 + 5.2^2 - 2(3.6)(5.2) \cos 68.8^\circ$   
 $= 26.461$  (to 5 s.f.)  
 $BD = \sqrt{26.461}$  ( $BD > 0$ )  
 $= 5.14$  cm (to 3 s.f.)

#### EXAM TIP

The Cosine Rule relates three sides and one angle.

- (c) Using Sine Rule,  

$$\frac{\sin \angle CBD}{CD} = \frac{\sin \angle BCD}{BD}$$

$$\frac{\sin \angle CBD}{2.1} = \frac{\sin 111.2^\circ}{5.1440}$$

$$\sin \angle CBD = \frac{2.1 \sin 111.2^\circ}{5.1440}$$

$$= 0.380\,61$$
 (to 5 s.f.)  
 $\angle CBD = \sin^{-1} 0.380\,61$   
 $= 22.372^\circ$  (to 3 d.p.)  
 $\therefore \angle BDC = 180^\circ - 111.2^\circ - 22.372^\circ$  ( $\angle$  sum of  $\triangle$ )  
 $= 46.4^\circ$  (to 1 d.p.)

#### EXAM TIP

The Sine Rule, which relates two sides and two angles, may be used to calculate  $\angle CBD$ , before using the property of the angle sum of a triangle to calculate  $\angle BDC$ .

12. (a) (i) Variety A has a smaller standard deviation, so the total mass of potatoes from the 5 plants will have a smaller spread.
- (ii) Variety B has a greater mean, indicating that on average, the total mass of potatoes from the 50 plants will be greater than if Feng were to choose variety A.

- (b) (i) Median = **29 cm**
- (ii) Upper quartile = 38 cm  
Lower quartile = 21 cm  
 $\therefore$  Interquartile range  
 $= (38 - 21)$  cm  
 $= 17$  cm

#### EXAM TIP

Recall that interquartile range = upper quartile – lower quartile.

- (iii) P(one has a height of less than 10 cm and the other has a height of more than 40 cm)  
 $= \frac{6}{120} \times \frac{24}{119} + \frac{24}{120} \times \frac{6}{119}$   
 $= \frac{12}{595}$

## Paper 1

September/October 2023

1. (a) 0.003 482 5 = **0.003 48** (to 3 s.f.)

#### EXAM TIP

In a decimal, all zeros before a non-zero digit are not significant.

- (b) 135% of \$24.60 =  $\frac{135}{100} \times \$24.60$   
 $= \$33.21$

2.  $4a - 2b - a + 3b = 3a + b$

#### EXAM TIP

Group the like terms.

3. (a) 0.000 078 6 =  **$7.86 \times 10^{-5}$**

#### EXAM TIP

A number is expressed in standard form when written as  $A \times 10^n$ , where  $n$  is an integer and  $1 \leq A < 10$ .

- (b)  $(6.3 \times 10^3)^2 = 39\,690\,000$   
 $= \mathbf{3.969 \times 10^7}$

4. (a) If there are 50 people in the group,  
Number who can ride a bicycle  
 $= \frac{90^\circ}{360^\circ} \times 50$   
 $= 12.5$ , which is not a positive integer  
 $\therefore$  There cannot be 50 people in the group. She is not correct.

#### EXAM TIP

Recall the angle of the sector is proportional to the number of people. Find the quantity that corresponds to  $90^\circ$  or  $198^\circ$  given that  $360^\circ$  represents 50 people.

- (b) Fraction of group who can drive a car only

$$= \frac{72^\circ}{360^\circ}$$

$$= \frac{1}{5}$$

Fraction of group who can ride a bicycle only

$$= \frac{90^\circ}{360^\circ}$$

$$= \frac{1}{4}$$

Fraction of group who can drive a car and ride a bicycle

$$= \frac{198^\circ}{360^\circ}$$

$$= \frac{11}{20}$$

LCM of 5, 4 and 20 = 20

$\therefore$  There could be 20 people in the group.

#### EXAM TIP

The number of people in the group can be any positive multiple of 20.

$$5. \quad AB^2 + BC^2 = 4^2 + 9.6^2$$

$$= 108.16$$

$$AC^2 = 10.4^2$$

$$= 108.16$$

Since  $AB^2 + BC^2 = AC^2$ , by the converse of Pythagoras' Theorem,  $\triangle ABC$  is a right-angled triangle.

$\therefore$  Lim is correct.

#### EXAM TIP

Determine if Pythagoras' Theorem holds true.

6. (a) A possible value of  $a$  is  $-1$ .

#### EXAM TIP

When the coefficient of  $x^2$  is negative, the graph opens downwards.

$$(b) \quad x = \frac{-3+7}{2} = 2$$

$\therefore$  Equation of line of symmetry:  $x = 2$

#### EXAM TIP

The graph cuts the  $x$ -axis when  $y = 0$  and the line of symmetry passes through the middle of these 2 points.

$$7. \quad x\text{-coordinate of } B = 1 + 5$$

$$= 6$$

$$y\text{-coordinate of } B = 4 - 3$$

$$= 1$$

$$\therefore B(6, 1)$$

#### EXAM TIP

$M$  is the midpoint of  $AB$ .

8. (a) Number of petals = 26  
(b) Mode = 4.8 cm

#### EXAM TIP

The mode is the value with the highest frequency.

$$9. \quad (a) \quad 4 - 2x = 5$$

$$-2x = 1$$

$$x = -\frac{1}{2}$$

$$(b) \quad y = \frac{k}{x^3} \quad \text{--- (1)}$$

Substitute  $y = 0.2$  and  $x = 2$  into (1):

$$0.2 = \frac{k}{2^3}$$

$$k = \frac{8}{5}$$

$$\therefore y = \frac{8}{5x^3}$$

When  $x = 0.5$ ,

$$y = \frac{8}{5(0.5)^3}$$

$$= 12.8$$

#### EXAM TIP

Substitute  $y = 0.2$  and  $x = 2$  into  $y = \frac{k}{x^3}$  to find the value of  $k$ . Then substitute  $x = 0.5$  to find  $y$ .

$$10. \quad (a) \quad \text{Distance travelled} = 42 \times \frac{50}{60}$$

$$= 35 \text{ km}$$

#### EXAM TIP

Convert 50 min into hours first.

$$(b) \quad 42 \text{ km/h} = \frac{(42 \times 1000) \text{ m}}{(60 \times 60) \text{ s}}$$

$$= 11\frac{2}{3} \text{ m/s}$$

$$11. \quad (a) \quad \left(\frac{3m}{2t^2}\right)\left(\frac{7mt}{6}\right) = \frac{7m^2}{4t}$$

#### EXAM TIP

Apply  $a^m \times a^n = a^{m+n}$  and  $a^m \div a^n = a^{m-n}$ .

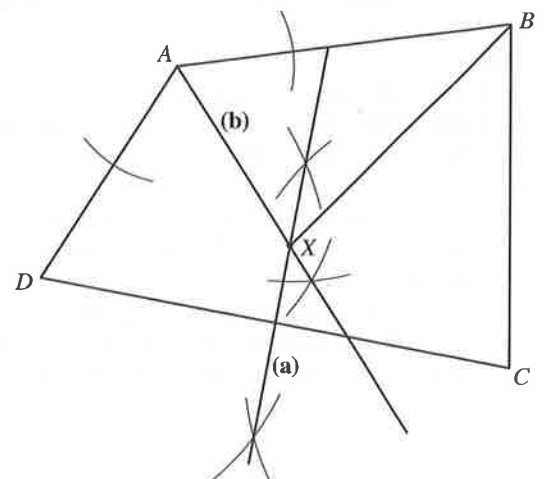
$$(b) \quad 4x < 18$$

$$x < 4.5$$

#### EXAM TIP

The inequality sign does not reverse when both sides are divided by a positive number.

12.



(c)  $\angle XBC = 45^\circ$

13. (a)  $\sqrt[3]{8000} = \sqrt[3]{2^6 \times 5^3}$   
 $= 2^2 \times 5$

(b)  $8000 = 2^6 \times 5^3$

$X = 2 \times 3^2 \times 5^4$

HCF of 8000 and  $X = 2 \times 5^3$

LCM of 8000 and  $X = 2^6 \times 3^2 \times 5^4$

$\therefore a = 1, b = 2, c = 4$

#### EXAM TIP

The HCF is the product of all the common prime factors of the two numbers.

The LCM is the product of all the common prime factors and all other prime factors of the two numbers.

14.  $x - 2y = 5$  — (1)

$5x + 6y = 1$  — (2)

From (1),

$x = 2y + 5$  — (3)

Substitute (3) into (2):

$5(2y + 5) + 6y = 1$

$10y + 25 + 6y = 1$

$16y = -24$

$y = -1.5$

Substitute  $y = -1.5$  into (3):

$x = 2(-1.5) + 5$

$= 2$

$\therefore x = 2, y = -1.5$

#### EXAM TIP

Apply the elimination or substitution method.

15. Angle  $DEB = 121^\circ$  because  $\angle ADE = \angle DCB = 54^\circ$   
 (corr.  $\angle$ s,  $DE \parallel CB$ ) and  $\angle DEB = \angle DAE + \angle ADE$   
 (ext.  $\angle$  of  $\triangle ADE$ ).

#### EXAM TIP

Identify the corresponding angle of the parallel lines  $DE$  and  $CB$  that is equal to  $54^\circ$ .

16.  $x^2 + 6x + 4 = 0$

$(x + 3)^2 - 9 + 4 = 0$

$(x + 3)^2 = 5$

$x + 3 = \pm\sqrt{5}$

$x = \sqrt{5} - 3$  or  $x = -\sqrt{5} - 3$

$= -0.76$  (to 2 d.p.)  $= -5.24$  (to 2 d.p.)

#### EXAM TIP

Solve  $x + a = \pm\sqrt{b}$ , as obtained from  $(x + a)^2 = b$ .

17. (a) Cheng's weekly pay =  $\frac{\$23\,400}{48} = \$487.50$

Cheng's daily pay =  $\frac{\$487.50}{5} = \$97.50$

Cheng's hourly pay =  $\frac{\$97.50}{6\frac{1}{4}} = \$15.60$

(b)  $6\frac{1}{4}$  h = 6 h 15 min

8.45 a.m. to 12.30 p.m.: 3 h 45 min

Amount of time Cheng works from 1.15 p.m.

= 2 h 30 min

$\therefore$  Cheng finishes work at **3.45 p.m.**

#### EXAM TIP

Write  $6\frac{1}{4}$  h as 6 h 15 min first.

18. (a) blue : red : yellow

3 : 4

6 : 7

9 : 12 : 14

$\therefore$  Blue : red : yellow = **9 : 12 : 14**

#### EXAM TIP

Find the LCM of 4 and 6.

(b) Number of yellow balls =  $\frac{54}{9} \times 14$   
 $= 84$

#### EXAM TIP

Alternatively, find the total number of balls in the box first.

19. (a) Actual length of the road =  $6.2 \text{ cm} \times 25\,000$   
 $= 155\,000 \text{ cm}$   
 $= \mathbf{1.55 \text{ km}}$

#### EXAM TIP

1 : 25 000 means 1 cm on the map represents 25 000 cm on the actual ground. Find the distance that 6.2 cm represents and convert the value to km.

(b) 25 000 cm is represented by 1 cm

0.25 km is represented by 1 cm

$0.25^2 \text{ km}^2$  is represented by  $1 \text{ cm}^2$

$\therefore$  Area of the field on the map =  $\frac{0.4}{0.25^2}$   
 $= \mathbf{6.4 \text{ cm}^2}$

#### EXAM TIP

25 000 cm is represented by 1 cm, so  $25\,000^2 \text{ cm}^2$  (or  $0.25^2 \text{ km}^2$ ) is represented by  $1 \text{ cm}^2$ .

20. Number of employees who took 20 min or less

$= 9 + 21$

$= 30$

Let the total number of employees be  $n$ .

$\frac{30}{n} = 0.4$

$n = 75$

$\therefore$  Number of employees in the last interval

$= 75 - 30 - 23 - 14$

$= 8$

#### EXAM TIP

$P(\text{an employee took 20 min or less}) =$   
 $\frac{\text{number of employees who took 20 min or less}}{\text{total number of employees}}$

21. (a)  $h = \frac{5}{4} (2r)$

$h = 2.5r$

(b) Volume  $= \pi r^2 h$

$364 = \pi r^2 (2.5r)$

$r^3 = \frac{728}{5\pi}$

$r = \sqrt[3]{\frac{728}{5\pi}}$

$= 3.59$  (to 3 s.f.)

**EXAM TIP**

Recall volume of cylinder  $= \pi r^2 h$ .

22.  $\frac{3}{x-2} + \frac{1}{x+4} = 5$

$\frac{3(x+4) + (x-2)}{(x-2)(x+4)} = 5$

$\frac{3x+12+x-2}{(x-2)(x+4)} = 5$

$4x+10 = 5(x-2)(x+4)$

$= 5(x^2+2x-8)$

$= 5x^2+10x-40$

$5x^2+6x-50=0$

$x = \frac{-6 \pm \sqrt{6^2 - 4(5)(-50)}}{2(5)}$

$= \frac{-6 \pm \sqrt{1036}}{10}$

$= 2.62$  or  $-3.82$  (to 2 d.p.)

**EXAM TIP**

Apply the quadratic formula:  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

23. (a) Arc length  $= \frac{75^\circ}{360^\circ} \times 2\pi(15)$

$= 6.25\pi$

$= 19.6$  cm (to 3 s.f.)

**EXAM TIP**

Recall arc length  $= \frac{\theta^\circ}{360^\circ} \times 2\pi r$ .

(b) Curved surface area of hemisphere  $= 200$  cm<sup>2</sup>

$\frac{1}{2} (4\pi r^2) = 200$

$2\pi r^2 = 200$

$r^2 = \frac{100}{\pi}$

$r = \sqrt{\frac{100}{\pi}} \quad (r > 0)$

$\therefore$  Curved surface area of cone  $= \pi r(8)$

$= 8\pi \sqrt{\frac{100}{\pi}}$

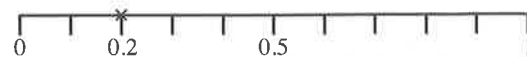
$= 142$  cm<sup>2</sup> (to 3 s.f.)

**EXAM TIP**

Recall curved surface area of a sphere  $= 4\pi r^2$ .

1. (a)  $\frac{-5 + \sqrt{(-5)^2 - 4 \times 3 \times (-7)}}{6} = 0.907$  (to 3 s.f.)

(b)  $\sqrt[3]{0.008} = 0.2$



(c)  $2\,889\,330$  mm  $= 2889.33$  m

$42\,300$  cm  $= 423$  m

$0.395$  km  $= 395$  m

$\therefore 380$  m,  $0.395$  km,  $42\,300$  cm,  $2\,889\,330$  mm

**EXAM TIP**

Convert all the lengths into the same unit.

2. (a)  $\frac{2^5 \times 4^3}{2^{-2}} = \frac{2^5 \times 2^6}{2^{-2}}$   
 $= 2^{5+6-(-2)}$   
 $= 2^{13}$

**EXAM TIP**

Apply  $a^m \times a^n = a^{m+n}$ ,  $a^m \div a^n = a^{m-n}$  and  $(a^m)^n = a^{mn}$ .

(b)  $\left(\frac{x^4}{9}\right)^{\frac{3}{2}} = \left(\frac{9}{x^4}\right)^{\frac{3}{2}}$   
 $= \left(\frac{3^2}{x^4}\right)^{\frac{3}{2}}$   
 $= \frac{3^3}{x^6}$   
 $= \frac{27}{x^6}$

**EXAM TIP**

Apply  $\left(\frac{x}{y}\right)^{-n} = \left(\frac{y}{x}\right)^n$ .

3. (a)  $13(0) + 10(1) + 9(2) + 7(3) + p(4) + 4(5) + 2(6) = 101$

$4p + 81 = 101$

$4p = 20$

$p = 5$  (shown)

(b) Median position  $= \frac{13+10+9+7+5+4+2+1}{2}$

$= 25.5$

Median  $= \frac{2+2}{2}$

$= 2$

**EXAM TIP**

If the number of terms is even, the median is the mean of the two middle values.

(c)  $P(\text{fewer than 2 visits}) = \frac{23}{50}$

4. (a) Interior angle of regular hexagon

$= \frac{(6-2) \times 180^\circ}{6}$

$= 120^\circ$

**EXAM TIP**

Sum of interior angles in a polygon =  $(n - 2) \times 180^\circ$

- (b) Divide the hexagon into 6 congruent equilateral triangles.

$$\begin{aligned}\text{Area of hexagon} &= 6 \times \frac{1}{2} (5)(5) \sin 60^\circ \\ &= \mathbf{65.0 \text{ cm}^2} \text{ (to 3 s.f.)}\end{aligned}$$

**EXAM TIP**

A regular hexagon may be divided into 6 congruent equilateral triangles.

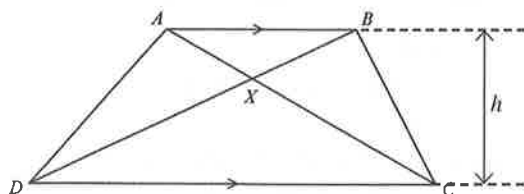
Apply area of triangle =  $\frac{1}{2} \times ab \times \sin C$ .

(c) Volume of pyramid =  $\frac{1}{3} \times 64.952 \times 12$   
 $= \mathbf{260 \text{ cm}^3}$  (to 3 s.f.)

**EXAM TIP**

Recall volume of pyramid =  $\frac{1}{3} \times \text{base area} \times \text{height}$

5.



(a) Area of  $\triangle ADC = \frac{1}{2} \times DC \times h$

$$\text{Area of } \triangle BDC = \frac{1}{2} \times DC \times h$$

$$\text{Area of } \triangle ADC = \text{Area of } \triangle BDC$$

$$\text{Area of } \triangle AXD + \text{Area of } \triangle XDC$$

$$= \text{Area of } \triangle BXC + \text{Area of } \triangle XDC$$

$$\therefore \text{Area of } \triangle AXD = \text{Area of } \triangle BXC \text{ (shown)}$$

- (b) Since  $\triangle AXB$  is similar to  $\triangle CXD$ ,

$$\frac{DX}{BX} = \frac{CX}{AX}$$

$$\frac{DX}{7.5} = \frac{8}{5}$$

$$DX = \frac{8}{5} \times 7.5$$

$$= 12 \text{ cm}$$

$$\therefore BD = BX + DX$$

$$= 7.5 + 12$$

$$= \mathbf{19.5 \text{ cm}}$$

**EXAM TIP**

Since the triangles are similar, their corresponding sides are in the same ratio.

6. (a) Using Pythagoras' Theorem,

$$AB^2 = 8^2 - 5^2$$

$$= 39$$

$$AB = \sqrt{39} \text{ (} AB > 0 \text{)}$$

$$= \mathbf{6.24 \text{ cm}} \text{ (to 3 s.f.)}$$

- (b) Consider  $\triangle BCD$ .

$$\tan 65^\circ = \frac{8}{CD}$$

$$CD = \frac{8}{\tan 65^\circ}$$

$$= 3.7305 \text{ cm (to 5 s.f.)}$$

$$\begin{aligned}\text{Area of } ABCD &= \frac{1}{2} \times \sqrt{39} \times 5 + \frac{1}{2} \times 8 \times 3.7305 \\ &= \mathbf{30.5 \text{ cm}^2} \text{ (to 3 s.f.)}\end{aligned}$$

**EXAM TIP**

$\triangle BCD$  is a right-angled triangle, so apply

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

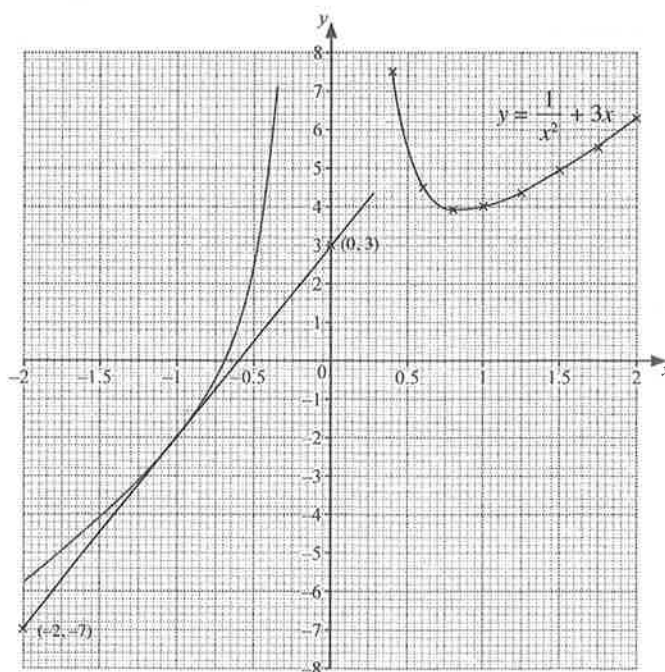
7. (a)  $y = \frac{1}{x^2} + 3x$

$$\text{When } x = 2,$$

$$y = \frac{1}{2^2} + 3(2)$$

$$= \mathbf{6.25}$$

(b)



- (c) When  $x = 0$ ,  $\frac{1}{x^2}$  is undefined. Therefore, it is not possible to find a value of  $y$  when  $x = 0$ .

- (d) When  $y = 5$ ,  $x = -0.4$ ,  $x = 0.55$  or  $x = 1.5$ .

(e) Gradient =  $\frac{3 - (-7)}{0 - (-2)}$   
 $= \mathbf{5}$

**EXAM TIP**

$$\text{Recall gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

8. (a) Basheera's age:  $x$

$$\text{Asif's age: } \frac{x}{2}$$

$$\text{Chen's age: } x + 10$$

$$\text{Given that mean} = 15,$$

$$\left(x + \frac{x}{2} + x + 10\right) \div 3 = 15$$

$$2.5x + 10 = 45$$

$$2.5x = 35$$

$$x = 14$$

$\therefore$  Asif is **7 years old**, Basheera is **14 years old** and Chen is **24 years old**.

(b)  $y = \frac{ax}{bx - c}$

$$y(bx - c) = ax$$

$$bxy - cy = ax$$

$$bxy - ax = cy$$

$$x(by - a) = cy$$

$$x = \frac{cy}{by - a}$$

#### EXAM TIP

To make  $x$  the subject of the formula, rearrange the formula such that  $x$  is alone on the LHS of the equation.

(c)  $6x^2 - x - 5 = (6x + 5)(x - 1)$

#### EXAM TIP

Use a multiplication frame to factorise the quadratic expression.

9. (a)  $x$ -coordinate of  $D = 9$

$$y\text{-coordinate of } D = 12 - 4 = 8$$

$$\therefore D(9, 8)$$

- (b) (i) Equation of  $AB$ :  $x = 3$

$$\begin{aligned} \text{(ii) Gradient of } AC &= \frac{12 - 2}{9 - 3} \\ &= \frac{5}{3} \end{aligned}$$

$$\text{Equation of } AC: y - 2 = \frac{5}{3}(x - 3)$$

$$= \frac{5}{3}x - 5$$

$$y = \frac{5}{3}x - 3$$

#### EXAM TIP

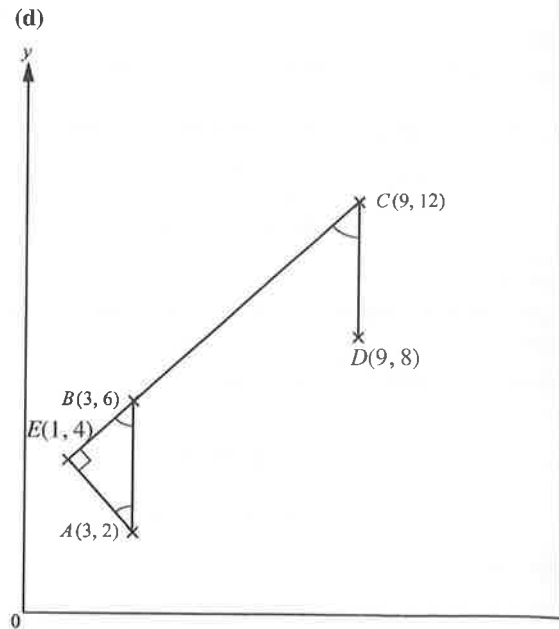
The equation of a line is in the form  $y = mx + c$ , where  $m$  is the gradient and  $c$  is the  $y$ -intercept.

(c) Area of  $ABCD = 4 \times 6$

$$= 24 \text{ square units}$$

#### EXAM TIP

Recall area of parallelogram = base  $\times$  height.



$$\begin{aligned} \angle EBA &= \frac{180^\circ - 90^\circ}{2} \text{ (base } \angle \text{ s of isos. } \triangle) \\ &= 45^\circ \end{aligned}$$

$$\begin{aligned} \angle BCD &= \angle EBA \text{ (corr. } \angle \text{ s, } CD \parallel BA) \\ &= 45^\circ \end{aligned}$$

10. (a) Difference =  $(390\,332 - 358\,975)$  million US\$  
= **31\,357 million US\$**

- (b) Value of exports in 2016

$$= \frac{100}{110.4001} \times 373\,255 \text{ million US$}$$

$$= \mathbf{338\,093 \text{ million US$}} \text{ (to the nearest million)}$$

#### EXAM TIP

Find 100% given that 110.4001% represents 373 255 million US\$

- (c) Estimated value of exports of plastics

$$= 3.76\% \times 15\,184 \text{ million US$}$$

$$= \mathbf{571 \text{ million US$}} \text{ (to the nearest million)}$$

- (d) Value of exports excluding plastics and mineral fuels

$$= \frac{100 - 12.2 - 3.76}{100} \times 390\,332 \text{ million US$}$$

$$= 328\,035 \text{ million US$ (to the nearest million)}$$

$$< 358\,975 \text{ million US$}$$

$\therefore$  Raffi is correct.

11. (a) (i) Median = **39 thousand hours**

- (ii) Upper quartile = 49 thousand hours

$$\text{Lower quartile} = 30 \text{ thousand hours}$$

$$\therefore \text{Interquartile range}$$

$$= (49 - 30) \text{ thousand hours}$$

$$= \mathbf{19 \text{ thousand hours}}$$

- (iii) Interquartile range of Brand B

$$= (53 - 26) \text{ thousand hours}$$

$$= 27 \text{ thousand hours}$$

The interquartile range of Brand B is greater than that of Brand A, so the lifetimes of light bulbs of Brand B are less consistent than those of Brand A.

#### EXAM TIP

Compare the interquartile range of both brands.

- (iv) Median of Brand B = 42 thousand hours

The median lifetime of light bulbs of Brand B is greater than those of Brand A, so the light bulbs of Brand A generally last longer than those of Brand B.

#### EXAM TIP

Compare the medians of both brands.

$$\begin{aligned} \text{(b) } P(\text{exactly 1 ball is red}) &= \frac{7}{12} \times \frac{5}{11} \times \frac{4}{10} \times 3 \\ &= \frac{7}{22} \end{aligned}$$

12. (a) (i) Angle  $AOC = 2.4$  rad

Reason: **Angle at centre = 2 × angle at circumference**

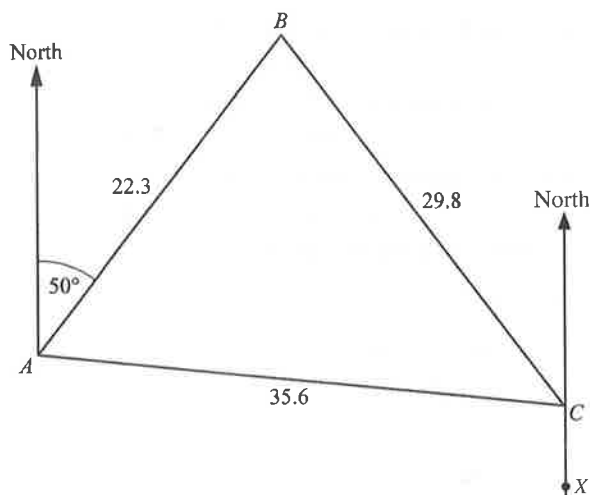
- (ii) Angle  $ADC = (\pi - 1.2) = 1.94$  rad (to 3 s.f.)

Reason: **Angles in opposite segments are supplementary**

#### EXAM TIP

The property of angles in opposite segments cannot be used for quadrilateral  $OADC$  because  $O$  does not lie on the circumference of the circle.

(b)



Using Cosine Rule,

$$29.8^2 = 22.3^2 + 35.6^2 - 2(22.3)(35.6) \cos \angle BAC$$

$$888.04 = 1764.65 - 1587.76 \cos \angle BAC$$

$$\cos \angle BAC = 0.55210 \text{ (to 5 s.f.)}$$

$$\angle BAC = \cos^{-1} 0.55210$$

$$= 56.488^\circ \text{ (to 3 d.p.)}$$

$$\angle ACX = 50^\circ + 56.488^\circ \text{ (alt. } \angle\text{s, } \parallel \text{ lines)}$$

$$= 106.488^\circ \text{ (to 3 d.p.)}$$

$$\therefore \text{Bearing of A from C} = 180^\circ + 106.488^\circ$$

$$= 286.5^\circ \text{ (to 1 d.p.)}$$

#### EXAM TIP

The bearing of A from C is the angle from a 'North' arrow from C measured clockwise to line AC.

$$\begin{aligned} 1. \quad \frac{285 - 62}{\sqrt{0.0896}} &\approx \frac{300 - 60}{\sqrt{0.09}} \\ &= \frac{240}{0.3} \\ &= \frac{2400}{3} \\ &= 800 \end{aligned}$$

#### EXAM TIP

Recall in a decimal, all zeros before a non-zero digit are not significant.

2. (a)  $p = 180^\circ - 134^\circ$  (adj.  $\angle$ s on a str. line, corr.  $\angle$ s)  
 $= 46^\circ$

#### EXAM TIP

Recall angle properties formed by parallel lines.

$$\begin{aligned} \text{(b) } q &= 75^\circ + 46^\circ \text{ (ext. } \angle \text{ of a } \triangle) \\ &= 121^\circ \end{aligned}$$

#### EXAM TIP

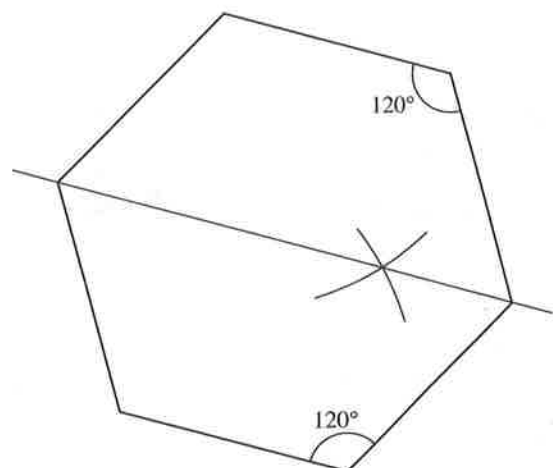
Recall exterior angle of a triangle is the sum of its interior opposite angles.

3. Price decrease =  $\$165 - \$138$   
 $= \$27$   
 Percentage decrease =  $\frac{\$27}{\$165} \times 100\%$   
 $= 16.4\% \text{ (to 3 s.f.)}$

#### EXAM TIP

$$\text{Percentage decrease} = \frac{\text{decrease}}{\text{original value}} \times 100\%$$

- 4.



#### EXAM TIP

Each interior angle of a regular hexagon is  $120^\circ$ .